

Notes: Zingales (RFS, 1994)

The Value of the Voting Right: A Study of the Milan Stock Exchange Experience

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1 Big picture

- **Question.** How much is the right to vote worth in common stock, and what determines that value?
- **Setting.** The paper studies Italian dual-class firms on the Milan Stock Exchange between 1987 and 1990. These firms have voting common shares and nonvoting “savings shares” that actually have better dividend rights.
- **Main fact.** Voting shares trade at an average premium of about 82% relative to nonvoting shares. After accounting for the extra dividend promised to nonvoting shares, the voting premium is above 90%.
- **Why this is surprising.** Previous studies in the United States, Sweden, Switzerland, England, Canada, and Israel usually find much smaller voting-right premia. Italy is an extreme case.
- **Core mechanism.** Voting rights are valuable because they allocate control. When private benefits of control are high and votes held by outside shareholders may be pivotal, outside votes command a large premium.
- **Ownership structure.** Italian firms have very concentrated ownership. The largest shareholder owns about 52% of votes on average, and many companies also have additional large shareholders.
- **Empirical prediction.** Voting premia should be lower when one shareholder already has majority control, and higher when small shareholders’ votes are strategically important in a control contest.
- **Main proxy.** Zingales uses the relative Shapley value (RSV) of small shareholders’ votes. This measures how pivotal those votes are relative to the fraction of votes held by small shareholders.
- **Regression result.** The voting premium is positively related to RSV divided by the fraction of voting shares. In the pooled regression, the coefficient is 29.2, implying very large private benefits of control.
- **Alternative proxies.** A majority-control dummy, the size of the largest shareholder, and the size of the second-largest shareholder all support the same story.
- **Private-benefits interpretation.** Even when control seems locked in, voting shares remain very valuable. The paper argues this reflects the market’s anticipation of future control changes and the huge value of control itself.
- **Case-study evidence.** The privatization news for Credito Italiano in 1992 generated an increase in the control premium of roughly 60% of the value of nonvoting equity.
- **Economic takeaway.** The paper provides early and influential evidence that private benefits of control can be enormous when minority investor protection is weak and the market for corporate control gives votes strategic value.

2 Data and measurement: voting shares, savings shares, and control rights

2.1 Why Italy is a useful laboratory

- Italian companies can issue ordinary voting shares and nonvoting savings shares.
- Multiple-vote shares were outlawed in 1942, so the comparison is clean: one class has ordinary voting rights, while the savings shares do not vote.
- Savings shares are not inferior cash-flow claims. They have minimum-dividend and extra-dividend privileges.
- Therefore, if voting shares trade above nonvoting shares, the premium cannot be explained by superior dividends attached to voting shares.

Interpretation. The Italian setting is powerful because voting shares have weaker dividend rights yet still trade at a large premium. The price difference is therefore a direct window into the value of control.

2.2 Savings shares and dividend privileges

- Savings shares, or *azioni di risparmio*, are entitled to a minimum dividend equal to 5% of par value.
- Whenever voting common shareholders receive a dividend, savings shares receive the same dividend plus at least 2% of par value.
- In liquidation, savings shares have seniority up to par value and then equal rights thereafter.
- These privileges mean the raw price gap understates the value of voting rights.

Interpretation. The nonvoting class is not a weak version of common stock. It has compensating cash-flow rights, so the fact that it trades far below voting stock is especially striking.

2.3 Raw voting premium

The raw proportional voting premium for firm i in year t is

$$VP_{it} = \frac{P_{it}^v - P_{it}^{nv}}{P_{it}^{nv}}, \quad (1)$$

where P_{it}^v is the price of a voting share and P_{it}^{nv} is the price of a nonvoting savings share.

Interpretation. If voting and nonvoting shares had identical cash-flow rights, this price gap would directly measure the value of a vote. Because nonvoting shares have better dividends, this raw measure is conservative.

2.4 Dividend-adjusted voting premium

The estimated voting premium subtracts the value of the savings shares' extra dividend privilege:

$$\widehat{VP}_{it} = VP_{it} - \frac{\varepsilon_{it}}{\rho P_{it}^{nv}}, \quad (2)$$

where ε_{it} is the extra dividend paid to nonvoting shares and ρ is the appropriate discount factor.

Interpretation. The adjustment makes the result stronger. Since nonvoting shares receive extra dividends, the raw price difference understates the pure value of the voting right.

2.5 Sample construction

- The sample includes all companies with both voting and nonvoting stock traded on the Milan Stock Exchange between 1987 and 1990.
- Convertible nonvoting shares are excluded until their conversion right expires.
- The spread of nonvoting shares became important only in the mid-1980s; by 1987 there were 64 dual-class firms and by 1990 there were 84.
- The total sample includes 301 firm-years, with 299 firm-years after requiring ownership-structure data.
- Ownership structure, shares outstanding, and dividend privileges are taken from *Il Taccuino dell'Azionista*.
- Prices are measured over the first five trading days of each year to match the ownership data dated December 31 of the previous year.

Interpretation. The timing is designed to line up ownership structure with contemporaneous market prices. Zingales avoids annual averaging of prices because ownership changes during the year are not observable.

2.6 Liquidity and other explanations

- Nonvoting shares often have higher turnover as a proportion of shares outstanding.
- Voting shares often have higher total value traded because there are more voting shares outstanding.
- Bid-ask spread evidence suggests voting shares are, if anything, slightly less liquid than nonvoting shares.
- Convertible nonvoting shares trade close to voting shares, suggesting the huge discount on ordinary nonvoting shares is not a generic liquidity discount.

Interpretation. Liquidity is not a convincing explanation for an 82% voting premium. The paper's central explanation is control value, not transaction costs.

3 Models and empirical specifications: equations plus interpretation

3.1 Private benefits of control

A firm's income is divided conceptually into two components:

- V : verifiable income that belongs equally to all shareholders;
- B : nonverifiable private benefits that accrue to those in control.

Define the ratio of private to verifiable benefits as

$$\beta = \frac{B}{V}. \quad (3)$$

Interpretation. The central economic object is not total firm value alone. It is how much of the firm's value can be diverted, enjoyed, or controlled by whoever holds corporate control.

3.2 If private benefits were distributed evenly across voting shares

If private benefits were evenly allocated among all voting shares, then

$$B = (RT)N_v = (P_v - P_{nv})N_v, \quad (4)$$

where RT is the value of one voting right and N_v is the number of voting shares.

Let π be the fraction of shares that vote. Then the voting premium would be

$$VP = \frac{P_v - P_{nv}}{P_{nv}} = \frac{B}{V} \frac{N}{N_v} = \frac{\beta}{\pi}. \quad (5)$$

Interpretation. If private benefits were spread evenly over all voting shares, fewer voting shares would mechanically imply a larger premium per voting share.

3.3 Strategic allocation of private benefits and RSV

In reality, private benefits are not distributed evenly. Votes are more valuable when they are pivotal in a control contest. The paper therefore uses

$$VP = \beta \left(\frac{RSV}{\pi} \right), \quad (6)$$

where RSV is the relative Shapley value of small shareholders' votes.

Interpretation. RSV measures voting power rather than voting quantity. Small shareholders' votes are worth little when one block already has majority control, but they become very valuable when they can swing control between competing blocs.

3.4 Baseline regression

The main empirical specification is

$$\widehat{VP}_{it} = \alpha + \beta \left(\frac{RSV}{\pi} \right)_{it} + \gamma \left(\frac{\varepsilon}{P_{nv}} \right)_{it} + u_{it}. \quad (7)$$

- $\widehat{VP}_{it}$ is the adjusted voting premium.
- RSV/π measures the strategic value of small shareholders' votes.
- ε/P_{nv} controls for the extra dividend privilege of nonvoting shares.
- The coefficient β is interpreted as a measure of private benefits relative to verifiable income.

Interpretation. A positive coefficient on RSV/π means that votes are priced more highly when they are more pivotal in control allocation.

3.5 Alternative ownership proxies

The paper checks whether similar results arise using simpler ownership variables:

- a dummy equal to one if one shareholder owns at least 50% of votes;
- the voting stake of the largest shareholder;
- the voting stake of the second-largest shareholder;
- combinations of these variables with RSV/π .

Interpretation. These variables proxy for the same idea. Majority control should reduce competition for votes. A large second shareholder can increase competition for control and make outside votes more valuable.

3.6 Positive intercept specification

Because the baseline regression has a large positive intercept, the paper considers

$$\widehat{VP}_{it} = \alpha + \beta_0 \left(\frac{1}{\pi} \right)_{it} + \beta_1 \left(\frac{RSV}{\pi} \right)_{it} + \gamma \left(\frac{\varepsilon}{P_{nv}} \right)_{it} + u_{it}. \quad (8)$$

Interpretation. The $1/\pi$ term captures the possibility that even apparently locked-in control situations may change in the future, especially if takeover-law changes make control blocks contestable. This helps explain why voting premia remain large even when RSV is near zero.

3.7 Control premium lower bound

The case-study lower bound for private benefits uses

$$\text{Control premium} = N_v(P_v - P_{nv}) \leq B, \quad (9)$$

where B is the acquirer's private benefit of control.

Interpretation. If an acquirer pays more for voting shares than the value of the nonvoting cash-flow claim, the difference must reflect private benefits of control. It is a lower bound because the seller may not extract the buyer's full surplus.

4 Identification and empirical design

4.1 What identifies the main cross-sectional effect

- The main variation comes from differences across firms and years in ownership concentration and the pivotal value of small shareholders' votes.
- The dependent variable is the voting premium between two traded claims on the same firm.
- This within-firm security comparison removes many firm-level valuation confounds because voting and nonvoting shares are claims on the same company.
- Dividend privileges are directly controlled for.

Interpretation. The design asks whether the price of a vote varies with the strategic value of that vote in allocating control.

4.2 Why dual-class shares are useful

- Voting and nonvoting shares trade simultaneously.
- The nonvoting shares have stronger dividend rights, so the raw premium is conservative.
- Both securities are exposed to the same firm cash flows.
- The main remaining difference is control rights.

Interpretation. This is why the Milan dual-class setting can estimate the value of voting rights more directly than ordinary single-class stock prices.

4.3 Addressing liquidity concerns

- The paper discusses turnover, transaction values, and bid-ask spread evidence.
- Voting shares do not appear more liquid in a way that could explain the premium.
- The price relation for convertible nonvoting shares also cuts against a liquidity-only explanation.

Interpretation. The voting premium is too large and too systematically related to ownership structure to be explained by trading frictions alone.

4.4 Why the Shapley value matters

- Simple voting shares are not equally valuable.
- A small outside block can be pivotal when two large shareholders are close to control.
- The Shapley value converts the ownership structure into a measure of pivotal voting power.
- The paper also shows that simpler proxies produce similar signs and conclusions.

Interpretation. The Shapley value is not essential because it is the only valid power index. It is useful because it formalizes the idea that voting value depends on coalitions and control contests.

4.5 Case-study credibility

- The Credito Italiano privatization news is a separate source of evidence.
- It does not rely on the Shapley-value proxy.
- The control premium jumps when the probability of a change in control rises.
- The increases imply private benefits above 60% of nonvoting equity value.

Interpretation. The case study validates the regression interpretation: control itself appears extremely valuable in Italy.

4.6 What the paper cannot fully prove

- Ownership structures are not randomly assigned.
- The exact private-benefit channel is not directly observed.
- The Shapley value is a proxy for voting power, not a literal bargaining model of every control contest.
- The Credito Italiano evidence is a case study, not a broad sample of control transfers.
- The paper proposes weak minority-investor protection as an explanation but does not estimate legal enforcement directly.

Interpretation. The evidence is strongest as a triangulation: huge premia, ownership-structure dependence, and a separate privatization episode all point to large private benefits of control.

5 Tables and figures

The original paper has four tables and two figures. They are grouped below to save space and to make the comparisons faster to read. All visuals are directly cropped from the uploaded paper.

5.1 Table 1 + Figure 1: sample facts and pivotal voting power

Read: Table 1:

- The average voting premium is 81.5%; the median is 73.7%.
- The nonvoting dividend privilege averages about 1.0%, so the raw premium understates the value of voting rights.
- The fraction of voting shares averages 0.74.
- The largest shareholder owns 51.7% of the votes on average; the second-largest owns 6.3%.
- Average market capitalization is 931 million dollars, showing that the sample is not only small firms.

Read: Figure 1:

- RSV is low when one large shareholder already has clear majority control.
- RSV becomes high when two large shareholders have sizable but nonmajority blocks and small shareholders' votes can decide control.
- The surface peaks where outside votes are most pivotal.

Economic message: The descriptive evidence establishes the puzzle: Italy has huge voting premia despite better dividends for nonvoting shares. Figure 1 shows how voting power depends on control contests, not simply the number of votes.

Table 1
Summary statistics

Variable	Mean	Median	SD	Min.	Max.
Voting premium (%)	81.5	73.7	64.9	-44.4	435.0
Differential div. (%)	1.0	0.8	0.9	0.1	6.1
Fraction voting shares (π)	0.74	0.74	0.14	0.50	0.99
Size largest shareholder (%)	51.7	52.0	16.7	5.2	92.0
Size second-largest shareholder (%)	6.3	0.0	8.2	0.0	43.2
RSV	0.2	0.0	0.4	0.0	3.6
E-P ratio (%)	5.6	5.6	4.2	0.0	22.3
Market cap. (million \$)	931	348	1866	12	17,899

The voting premium is the grand average of VP_{it} , where VP_{it} is the percentage premium of voting over nonvoting shares for company i at the beginning (first five trading days) of year t . The dividend privilege is the percentage size of the additional dividend yield of the nonvoting shares at the beginning of each year. π is the fraction of voting shares over total number of shares in company i at the beginning of each year. The size of the largest (second largest) shareholder is computed as the percentage of the voting shares held at the beginning of each year by the largest (second largest) shareholder. RSV is the Shapley value of votes held by small shareholders (those who own less than 5 percent of the votes) divided by the fraction of votes they hold. Earnings to price ratios are 1990 data computed using voting shares prices. Market capitalization is in million U.S.\$ (1\$ = 1500 Lit.). Total number of observations 299.

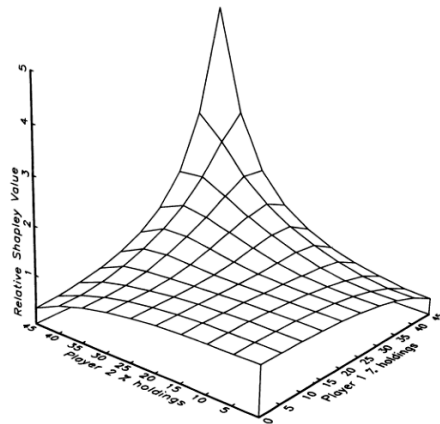


Table 1: Summary statistics

Figure 1: Relative Shapley value of small shareholders

5.2 Table 2 + Table 3: regression evidence on control value

Read: Table 2:

- In the pooled regression, the coefficient on RSV/π is 29.2 with a robust standard error of 7.5.
- The coefficient is positive in every year: 24.5 in 1987, 16.1 in 1988, 36.8 in 1989, and 35.2 in 1990.

- The dividend privilege coefficient is negative, as expected: a larger dividend advantage for nonvoting shares lowers the measured voting premium.
- The pooled R^2 is 9.3%, meaningful given the noisy firm-level ownership data and the simplicity of the specification.

Read: Table 3:

- The majority-owned dummy is negative: -30.2 when used alone.
- The size of the largest shareholder is negative: larger controlling blocks reduce the value of outside votes.
- The size of the second-largest shareholder is positive: a stronger rival block makes outside votes more valuable.
- Results are weaker in joint specifications because the ownership proxies are highly collinear.

Economic message: The regressions show that voting premia are tied to the allocation of control. Votes are worth more when they may be pivotal and less when majority control is already locked in.

Table 2
Determinants of the voting premium $VP_{it} = \alpha + \beta(RSV/\pi)_{it} + \gamma Privilege_{it} + \epsilon_{it}$

	1987	1988	1989	1990	Pooled
Constant	59.2 (6.0)	89.2 (16.8)	102.9 (12.1)	85.9 (9.5)	82.7 (7.7)
RSV/ π	24.5 (7.3)	16.1 (13.8)	36.8 (15.1)	35.2 (8.9)	29.2 (7.5)
Privilege	-4.8 (2.9)	-12.7 (6.5)	-11.7 (5.9)	-21.3 (7.8)	-10.3 (3.7)
R^2 (%)	16.7	5.2	9.6	19.6	9.3
Observations	62	75	78	84	299

VP_{it} is the percentage premium of voting over nonvoting shares for company i at the beginning (first five trading days) of year t . RSV_{it} is the Shapley value of votes held by small shareholders (those who own less than 5 percent of the votes) divided by the fraction of votes they hold. Votes ownership is computed at the beginning of each year, using data updated through the end of the previous year. π is the fraction of voting shares over total number of shares in company i at the beginning of each year. The privilege is the percentage size of the additional dividend yield of the nonvoting shares at the beginning of each year. All estimates are obtained by OLS. Heteroskedasticity robust standard errors are reported in parentheses. For the pooled regression the standard errors are robust with regard to heteroskedasticity and to serial correlation between observations of the same companies.

Table 3
Alternative specifications

	I	II	III	IV	V	VI	VII
Constant	82.3 (8.0)	112.5 (10.1)	92.4 (16.6)	131.5 (14.5)	89.3 (31.5)	83.5 (7.9)	79.5 (7.9)
Privilege	-10.2 (3.8)	-11.2 (3.8)	-10.5 (3.7)	-10.3 (3.8)	-10.2 (3.8)	-10.9 (3.9)	-10.2 (3.8)
RSV/ π	30.5 (9.9)		21.7 (16.6)		27.4 (19.6)		25.4 (11.2)
Majority-owned dummy		-30.2 (10.6)	-10.8 (17.0)				
Size largest shareholder				-0.77 (0.28)	-0.12 (0.54)		
Size second-larg. share-holder						1.42 (0.53)	0.70 (0.55)
R^2 (%)	8.3	7.7	8.4	6.7	8.3	5.8	8.9

RSV_{it} is the Shapley value of votes held by small shareholders (those who own less than 5 percent

Table 2: Determinants of the voting premium

Table 3: Alternative ownership specifications

5.3 Table 4 + Figure 2: the positive intercept and the Credito Italiano case

Read: Table 4:

- Column I repeats the baseline result after dropping one outlier.
- Column II adds inverse percentage voting shares. The constant becomes negative and insignificant, while inverse voting shares is positive and significant.
- The RSV/π coefficient remains positive and significant in columns II and III.
- Column III adds a state-owned dummy multiplied by inverse voting shares. The coefficient is -43.6 and highly significant.
- The R^2 rises from 8.3% in column I to 25.0% in column III.

Read: Figure 2:

- The control premium for Credito Italiano is roughly flat before privatization news.
- After rumors and the September 1992 official privatization announcement, the control premium jumps sharply.

- A second jump follows the November 1992 IRI board approval to sell the entire stake.
- The two updates together imply an increase of roughly 60% of the value of nonvoting equity.

Economic message: Table 4 explains why even firms with locked-in control can have large voting premia: future changes in control may matter. Figure 2 provides direct case-study evidence that private benefits of control in Italy were extremely large.

Table 4
The sources of a positive intercept

	I	II	III
Constant	82.3 (8.0)	-13.9 (26.2)	-25.7 (24.5)
Privilege	-10.2 (3.8)	-10.4 (3.7)	-10.7 (3.7)
Inverse perc. voting shares		68.4 (17.4)	82.7 (16.2)
RSV/ π	30.5 (9.9)	31.4 (8.5)	24.4 (8.2)
State-owned dummy			-43.6 (4.7)
R^2 (%)	8.3	16.8	25.0

RSV_{*it*} is the Shapley value of votes held by small shareholders (those who own less than 5 percent). Votes ownership is computed at the beginning of each year, using data updated through the end of the previous year. π is the fraction of voting shares in company at the beginning of each year.

Table 4: Sources of a positive intercept

Control Premium in Credito Italiano

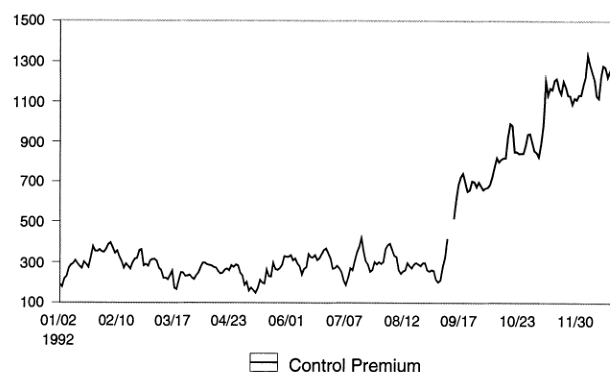


Figure 2

Control premium in Credito Italiano

The control premium is expressed in million U.S.\$ (1\$ = 1500 Lit.) and is computed as

$$\text{Control premium} = (P_v - P_{nv})N_v$$

where P_v (P_{nv}) is the price of a voting (nonvoting) share and N_v is the number of voting shares. The Italian government announced the decision to sell Credito Italiano to private investors on September 9, 1992. The decision to sell the whole stake owned by the government was approved by the board of directors of IRI, the state holding company controlling Credito Italiano, on November 10, 1992.

Figure 2: Control premium in Credito Italiano

6 Short summary

- Zingales studies the price of voting rights using Italian dual-class firms.
- The key comparison is voting common shares versus nonvoting savings shares.
- Savings shares have better dividend rights, so the voting-share premium is not compensation for better cash flows.
- The average raw voting premium is 81.5%, and the dividend-adjusted premium is above 90%.
- The paper interprets this premium as evidence of very large private benefits of control.
- The main ownership variable is the relative Shapley value of small shareholders' votes.
- RSV is high when small shareholders are likely to be pivotal in a control contest.
- Voting premia rise with RSV divided by the fraction of voting shares.
- Majority-owned firms have lower voting premia, consistent with weaker competition for control.
- A larger second shareholder raises the premium because it increases potential competition for control.
- A large intercept remains in the baseline model, suggesting that private benefits are valuable even when current control is stable.
- The paper explains the intercept by the possibility of future control changes and takeover-law changes.
- State-owned companies have lower premia, consistent with lower expected probability of control change during the sample period.
- The Credito Italiano privatization case provides separate evidence that private benefits exceed 60% of the value of nonvoting equity.
- The paper argues that weak protection of minority shareholders can explain why private benefits are so large in Italy.
- The broader implication is that the value of votes depends on both the size of private benefits and the probability that votes affect control.

7 Conclusion

7.1 Core conclusion

Zingales shows that voting rights can be extraordinarily valuable when corporate control provides large private benefits. In Italy, voting shares traded at a premium comparable to the value of dividend rights, even though nonvoting shares had superior dividend privileges. Ownership structure explains part of this premium: votes are most valuable when they can affect control.

7.2 Economic intuition

A vote is not valuable because it pays dividends. It is valuable because it can help allocate control. If the controller can extract private benefits, then votes that determine control become claims on those benefits. Small shareholders' votes become especially valuable when they are pivotal between competing large shareholders.

7.3 Takeaway

The paper is a classic demonstration that investor protection, private benefits, and ownership concentration shape security prices. The price of a vote is not a tiny governance detail; in weak-control environments it can be worth as much as the cash-flow claim itself.